

# Project 2 — Give the Model a Voice

Post-train the frozen Project 1 checkpoint to produce stories with a recognisable target narrative style, and prove the change without hiding a regression.

**Due:** 25 April 2027 · **Weight:** 50%

## Page summary

Starting from either your frozen Project 1 submission or the course narrative fallback checkpoint, post-train the model to produce stories with a recognisable target narrative style. The supplied target corpora are processed, verified public-domain editions of *Grimm's Fairy Tales* and *One Thousand and One Nights*. You may use one or both, or justify a permitted supplement or replacement.

“Post-training” is used broadly here: any training performed after Project 1 to change the model’s domain, style, or conditioned behaviour. You may choose target-domain continued pre-training, supervised instruction tuning, preference-based optimisation, or a justified combination. The method is open; the target behaviour and evaluation obligation are not.

## Key facts

| Item   | Value |
|--------|-------|
| Weight | 50%   |

| Item            | Value                                                              |
|-----------------|--------------------------------------------------------------------|
| Marks available | 50                                                                 |
| Training stage  | Post-training                                                      |
| Work mode       | Individual                                                         |
| Model size      | 32M design target; 33.6M absolute maximum                          |
| Compute budget  | 18 T4-equivalent GPU-hours and $2.5 \times 10^{17}$ training FLOPs |
| Main report     | 20 CVPR pages, excluding references and appendix                   |
| Due             | Sunday 25 April 2027, 23:59 AET — end of Week 8                    |

## What you must demonstrate

Define the target style in operational terms, use the supplied corpora or justify a permitted replacement, compare the post-trained model with the unchanged starting checkpoint, and evaluate target adaptation alongside narrative quality and retention. The project rewards the engineering fit between method, data, budget, and evidence — not the complexity or novelty of the post-training algorithm.

If you use instruction tuning, document the schema, special tokens, and loss masking. If you use preference optimisation, first establish an SFT or otherwise appropriate baseline and demonstrate why the extra pipeline and compute were justified. Work without genuine human preference labels must not be described as full RLHF.

## Required investigation

- define the target style in operational rather than purely impressionistic terms
- construct or curate suitable post-training data
- compare the post-trained model with its unchanged starting checkpoint
- separate held-out target-style texts at story or source level
- evaluate both target-style adaptation and narrative quality
- test for regressions in the original narrative capability
- analyse the effect of important training and sampling choices
- explain why the final method was selected over feasible alternatives

## Marking breakdown

| Component          | Marks     |
|--------------------|-----------|
| Engineering report | 35        |
| Submitted model    | 15        |
| <b>Total</b>       | <b>50</b> |

### Engineering report — 35 marks

**1. Target behaviour and method rationale — 5 marks.** A testable definition of the intended voice or style, success criteria beyond a vague resemblance claim, and a reasoned choice among continued pre-training, SFT, preference optimisation, or a combination.

**2. Data design and provenance — 5 marks.** Permitted and well-documented source editions, story/source-level splits, appropriate cleaning and formatting, token and source statistics, deduplication or overlap checks, and a defensible balance between target-style data and retained narrative ability.

**3. Implementation and reproducibility — 5 marks.** A correct transition from the starting checkpoint, a reproducible post-training pipeline, traceable

tokenizer/chat-template/loss-mask behaviour, declared hyperparameters and seeds, and a final package matching the reported method.

**4. Experimental design and budget judgement — 7 marks.** Meaningful baseline comparisons, controlled ablations or pilots, justified training duration and checkpoint selection, and evidence of prioritising experiments under the compute deadline.

**5. Multi-objective evaluation and regression analysis — 8 marks.** Held-out target-style language modelling, blind or clearly structured behavioural comparison, narrative coherence/completeness assessment, repetition and source-overlap checks, and evaluation of the original Project 1 capability.

**6. Communication, limitations, and engineering reflection — 5 marks.** A coherent report making the decision trail visible: initial expectation, evidence, revision, final choice, limitations, and what the project reveals about post-training under constraints.

## Submitted model — 15 marks

Submission validity is a technical gate. A valid model receives up to **10 marks** from the shared tutor-evaluation PPL rule and up to **5 marks** from blinded task-specific review.

**7. Tutor-evaluation perplexity — 10 marks.** The shared formula applied to continuation-only reference-token-normalised PPL across all ten tutor prompts:  $P \leq 25$  earns all 10 marks;  $25 < P < 50$  receives the published exponential decay;  $P \geq 50$  earns zero for this criterion.

**8. Target behaviour, narrative quality, and retention — 5 marks.** Whether readers can consistently recognise the operationally defined target behaviour across varied openings while the output remains fluent, connected, and narratively usable. Task-specific dimension weights: recognisable target behaviour **35%**; narrative coherence and usability **30%**; robustness across varied openings **20%**; retention and integrity, including unacceptable regression or

copying **15%**. A single showcase, memorised passage, or strong style signal attached to an incoherent story cannot earn high marks.

## Individual work

This project is submitted individually. AI assistance is permitted when declared in the required AI Assistance Statement; responsibility for the submitted work remains with you. See [Policies](#) for the full individual-work, collaboration, and AI-assistance rules.

## The spec

The brief poses the problem and leaves room for a response. The spec is the fixed part: what a marker checks. Some lines can be checked mechanically; the rest call for judgement.

- the starting checkpoint (own Project 1 submission or the course fallback) is declared and verified
- the target style is defined in operational rather than purely impressionistic terms
- the post-trained model is compared with its unchanged starting checkpoint on the same held-out cases
- held-out target-style texts are separated from training data at story or source level
- the report tests for regressions in the original Project 1 narrative capability
- any instruction schema, special tokens, and response-only loss masking are documented
- work without genuine human preference labels is not described as full RLHF
- training and evaluation stay within 18 T4-equivalent GPU-hours and  $2.5 \times 10^{17}$  training FLOPs

# Resources

## START HERE

### Full Project Brief

Printable copy of the complete task, constraints, evaluation rules, and rubric.

Will be available at 22 March 2027 at 9:00 am AET

### Project 2 Starter Repository

Assigned post-training code, starting-checkpoint loader, budget ledger, schemas, and submission manifest.

Will be available at 22 March 2027 at 9:00 am AET

## DATA AND MODEL

### Target-Style Corpus and Data Card

Versioned public-domain Grimm and One Thousand and One Nights data with provenance, splits, and checksums.

Will be available at 22 March 2027 at 9:00 am AET

### Post-training Starter Pack

Supported Colab and compact recipes for continued pre-training, SFT, masking inspection, and a toy preference exercise.

Will be available at 22 March 2027 at 9:00 am AET

## EVALUATE

### Behaviour Evaluation Pack

Five development examples, ten tutor prompts, separation validator, blind-review anchors, and retention checks.

Will be available at 22 March 2027 at 9:00 am AET

## SUBMIT

### CVPR Report Template

Shared two-column LaTeX template with the required engineering-report and AI Assistance Statement sections.

Will be available at 22 March 2027 at 9:00 am AET

## Submission

Will be available at 22 March 2027 at 12:00 am AET.

**Due:** 25 April 2027 at 11:59 pm AET

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## RELATED

[Project 1 — Build a Narrative Base Model](#)

[Project 3 — Build a Specialist](#)